Palindromic perfect powers in Human Friendly Form

A power saving over the Cilleruelo–Luca–Shparlinski bound

Lean-friendly version, matched block by block to the Lean formalization

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Overview

The theorem. Fix a base $g \geq 2$ and an exponent $k \geq 2$, and consider the palindromes with L digits in base g; there are about $g^{L/2}$ of them. Cilleruelo, Luca and Shparlinski [CLS] proved that at most a constant times $(\#\mathcal{P}_L)^{1/k}$ of them are perfect k-th powers. We prove (Theorem 101) that the exponent $1/k$ can be replaced by γ/k for some $\gamma = \gamma(g,k) < 1$, for every base and every exponent. This is a weak form of the main result of the accompanying paper [Pal], which gives explicit values such as $\gamma = 0.9674$ for $g = 2$; those values need a certified computer calculation, while the argument here needs none.

The starting point. If x^k is a palindrome with L digits, its first t digits are the reversal of its last t digits. We sort the candidates x by their residue class b modulo g^t, that is, by the last t digits of x (a *node*). The last t digits of x^k depend only on b, hence so do the first t digits of x^k, and these confine x to a short interval. So the possible x in a node form an arithmetic progression of length about $Z = g^{L/k} g^{-2t}$ (Lemmas 33 and 35), and counting all of them gives the bound of [CLS].

The strategy. We also use the next s digits from both ends. Inside a node write $x = x_b + g^t z$. The digits of x^k in positions $t, \dots, t+s-1$ form an affine function of z modulo g^s (Lemma 41), and the digits in positions $L-t-s, \dots, L-t-1$ form the integer part of a function of z that lies within $1/g$ of an affine function over the reals (Lemmas 42 and 44). The palindrome condition says that the second block is the reversal of the first (Lemma 45). We detect this with a Fejér-kernel majorant (Lemmas 50 and 51) and expand the reversal map in additive characters modulo g^s (Lemma 12). The coefficients of this expansion are the values of the *reversal transform* $\Psi_\ell(a)$, a discrete Fourier transform of digit reversal. Summing the resulting geometric series in z splits the count into a main term, smaller than the bound of [CLS] by the factor g^s, and an error term (Lemma 55).

The error term. Two ingredients control the error term. First, a power saving for the sum over ℓ of $\max_a |\Psi_\ell(a)|$ (Lemma 26). The modulus of $\Psi_\ell(a)$ is a product of digit factors (Lemma 11), and consecutive factors satisfy an identity in which a cancels (Lemma 16). So whenever two neighbouring digits of $(g^2-1)\ell \bmod g^s$ are not both 0 or both $g-1$, one of two consecutive factors is bounded away from 1, whatever a is (Lemma 22), and a count of digit strings (Lemma 24)

gives the saving. Arguments of this kind go back to Col [Col] and appear in [TP], [DMRSS] and [BhS]. Second, after replacing $|\Psi_\ell(a)|$ by its maximum over a, the sums over a are taken node by node (Lemma 59), and the averaging over the nodes comes from the archimedean phases alone: they are close to the values of a smooth monotone function of the first t digits, these values are well spaced (Lemmas 78 and 81), and few nodes share the same first digits (Lemma 68). The last point rests on an elementary count of k-th roots modulo prime powers (Lemmas 63–66) and on the g-part of a node, its largest divisor composed of primes dividing g (Lemma 67).

Large g-parts and parameters. Integers x that share a large factor with a power of g are counted separately by a direct argument (Lemma 89); this matters only when g is not a prime power. Finally we choose t with $g^{(4-c)t}$ about $g^{L/k}$, $s = t - 2$, and a threshold $D = g^{\eta L/k}$ for the g-part (Definition 90), and the bookkeeping (blocks 90–100) gives the theorem (Theorem 101).

Routine and delicate steps. Routine: digits, reversal and harmonic sums (blocks 1–8); the elementary properties of the reversal transform (blocks 9–15); the tree of nodes (blocks 28–36); the linearisation (blocks 37–45); the Fejér majorant (blocks 46–55); the sums over residues (blocks 56–59); the count of roots (blocks 60–66); the reduction to the quantities Φ_b (blocks 70–73); and the final bookkeeping with the deduction of the theorem (blocks 90–101). Delicate: the power saving (blocks 16–26); the rigidity of g-parts and the fibre bound (Lemmas 67 and 68); the replacement of the phases δ_b by values of a smooth function, and the spacing argument (blocks 74–86); and the large g-part (blocks 87–89). The key lemmas are Lemmas 26, 55, 68, 86, 89 and 100.

Prerequisites. Division with remainder, congruences, greatest common divisors, p-adic valuations and the Chinese remainder theorem; the binomial theorem; the mean value theorem, Taylor's theorem with the Lagrange remainder, the fact that a continuous function on a compact interval attains its maximum, and $\int_1^x \frac{dy}{y} = \log x$; the identity $|e(y) - 1| = 2|\sin \pi y|$ and the inequalities $|\sin \pi y| \geq 2\|y\|$, $|\sin y| \leq |y|$ and $\sin y \geq 2y/\pi$ for $0 \leq y \leq \pi/2$; and the properties $\|x - y\| \leq \|x\| + \|y\|$ and $\|nx\| \leq |n|\,\|x\|$ ($n \in \mathbb{Z}$) of the distance to the nearest integer. No result from the literature is imported, and no computer calculation is used. In the Lean formalization the algebraic prerequisites are taken from Lean and Mathlib, and the analytic ones are the fields of the structure `KnownResults` in `PalindromicPowers/Assumptions.lean`, documented in `palindromic_powers_lean/assumptions.md` and proved from Mathlib in `PalindromicPowers/Discharge.lean`; no published result is assumed.

About this document. This document follows the Human Friendly Form of [HFF]. After the mark ✠ the proof consists only of definitions, lemmas and remarks, numbered in one sequence; every lemma is proved directly below its statement in fewer than 300 visible characters (the length is shown in grey at the end of each proof); every block cites, by number, each earlier block whose notation or results it uses; and remarks carry no logical weight. This is the *Lean-friendly version*: its statements are those of the Lean formalization `palindromic_powers_lean`, and every definition, lemma and theorem names its Lean counterpart in a grey line below the statement (the Lean declarations of block n have docstrings beginning `[HFF n]`). The correspondence follows fixed conventions: the digit n_i is `digit g n i`; $\min(A, H/y)$ is `minDiv A H y`, which is A for $y = 0$ (the convention $H/0 = \infty$); $\|x\|$ is `distZ x`; and all constants are explicit: "$A \ll B$ for L large" becomes $\exists\, C_1, L_0\ \forall L \geq L_0 :\ A \leq C_1 B$, and ε_*, u_*, c,

C, t' and the parameters of Definition 90 are given by formulas. A table of notation is in `hff/LEAN_CONVENTIONS.md`. The dependency graph can be explored in the accompanying map, `palindromic_powers_hff_lean_map.html`. The document was written by Claude (Anthropic), from the accompanying paper [Pal], and revised to match the Lean statements.

✠

Definition 1 (Conventions)**.** Throughout, $g \geq 2$ and $k \geq 2$ are fixed integers. We write $e(x) = e^{2\pi i x}$, $\|x\|$ for the distance from $x \in \mathbb{R}$ to the nearest integer, log for the natural logarithm, and $n \bmod q \in \{0, \dots, q-1\}$ for the remainder of $n \in \mathbb{Z}$ on division by an integer $q \geq 1$. For real A, H, y, $\min(A, H/y)$ has its usual meaning if $y \neq 0$, and $\min(A, H/0) := A$ (the convention $H/0 = \infty$). "For L large" means for all $L \geq L_0$, where L_0 depends only on g and k; and "$A \ll B$ for L large" means: there are C_1 and L_0, depending only on g and k, with $|A| \leq C_1 B$ for all $L \geq L_0$.
Lean: `e`, `distZ`, `minDiv`

Lemma 2 (Digits)**.** *For $j \geq 0$, the map $(n_0, \dots, n_{j-1}) \mapsto \sum_{i<j} n_i g^i$ is a bijection from $\{0, \dots, g-1\}^j$ onto $\{0, \dots, g^j - 1\}$ (Definition 1).*
Lean: `digits_bijection`

Proof. The image lies in $[0, (g-1)\sum_{i<j} g^i] = [0, g^j - 1]$, and both sets have g^j elements, so it suffices to show injectivity. If $\sum_i n_i g^i = \sum_i n_i' g^i$, reducing modulo g gives $n_0 = n_0'$; subtract, divide by g, and use induction on j. [length 174/300] □

Definition 3 (Digits and reversal)**.** For integers $n, i \geq 0$, the i-th *digit* of n is $n_i := \lfloor n/g^i \rfloor \bmod g \in \{0, \dots, g-1\}$. For $j \geq 0$ put $\mathrm{rev}_j(n) := \sum_{i<j} n_i g^{j-1-i}$ (Definition 1).
Lean: `digit`, `rev`

Lemma 4 (Blocks of digits)**.** *Let $n \geq 0$ have digits n_r, and let $0 \leq i \leq i'$ (Definition 3). Then $n \bmod g^i = \sum_{r<i} n_r g^r$ and $\lfloor n/g^i \rfloor \bmod g^{i'-i} = \sum_{i \leq r < i'} n_r g^{r-i}$ (Definition 1). Moreover, if $j \geq 0$ and $d_r \in \{0, \dots, g-1\}$ for $r < j$, then the digits of $\sum_{r<j} d_r g^r$ are $d_0, \dots, d_{j-1}$ (in positions $0, \dots, j-1$).*
Lean: `digit_window`

Proof. The first claim follows by induction on i from $n \bmod g^{i+1} = n \bmod g^i + g^i n_i$. The second is the first for $\lfloor n/g^i \rfloor$, whose r-th digit is n_{i+r}, as $\lfloor \lfloor n/g^i \rfloor / g^r \rfloor = \lfloor n/g^{i+r} \rfloor$. For the third, $m := \sum_{r<j} d_r g^r < g^j$ (Lemma 2), so $m = m \bmod g^j = \sum_{r<j} m_r g^r$; now use the uniqueness in Lemma 2. [length 213/300] □

Lemma 5 (Reversal is an involution)**.** *Let $j \geq 0$. For every $n \geq 0$, $\mathrm{rev}_j(n) < g^j$ and the digits of $\mathrm{rev}_j(n)$ in positions $i < j$ are n_{j-1-i}; and $\mathrm{rev}_j(\mathrm{rev}_j(n)) = n$ if $n < g^j$ (Definitions 1 and 3).*
Lean: `rev_involution`

Proof. As $\mathrm{rev}_j(n) = \sum_{i<j} n_{j-1-i} g^i$ with all $n_{j-1-i} \leq g-1$, $\mathrm{rev}_j(n) < g^j$ by Lemma 2, and its digits are the n_{j-1-i} by Lemma 4. So $\mathrm{rev}_j(\mathrm{rev}_j(n)) = \sum_{i<j} n_i g^i = n \bmod g^j$ (Lemma 4), which is n if $n < g^j$. [length 145/300] □

Lemma 6 (Geometric sums)**.** *For $\theta \in \mathbb{R}$ and an integer $n \geq 1$, $\left|\sum_{z=0}^{n-1} e(z\theta)\right| \leq \min\left(n, \frac{1}{2\|\theta\|}\right)$ (Definition 1).*
Lean: `geom_sum_le`

Proof. Each term has modulus 1. If $\theta \notin \mathbb{Z}$, the sum is $(e(n\theta)-1)/(e(\theta)-1)$, of modulus at most $2/|e(\theta)-1| = 1/|\sin \pi\theta| \le 1/(2\|\theta\|)$, since $|\sin \pi\theta| \ge 2\|\theta\|$. [length 117/300] □

Lemma 7 (Harmonic numbers)**.** *For every integer $n \ge 1$, $\sum_{j=1}^{n} \frac{1}{j} \le 1 + \log n$ (Definition 1).*
Lean: `harmonic_le`

Proof. For $j \ge 2$, $\frac{1}{j} \le \int_{j-1}^{j} \frac{dx}{x}$, as $\frac{1}{x} \ge \frac{1}{j}$ on $[j-1, j]$. Summing over $2 \le j \le n$ gives $\sum_{j=1}^{n} \frac{1}{j} \le 1 + \int_1^n \frac{dx}{x} = 1 + \log n$. [length 78/300] □

Lemma 8 (An odd harmonic sum)**.** *For every integer $R \ge 1$, $\sum_{0<|j|\le R/2} \frac{1}{2|j|-1} \le 2 + \log R$ (Definition 1).*
Lean: `odd_harmonic_le`

Proof. For $R = 1$ the sum is empty. For $R \ge 2$ it is $2\sum_{1\le j\le R/2} \frac{1}{2j-1} \le 2 + 2\int_1^{R/2} \frac{dx}{2x-1} = 2 + \log(R-1)$, since $\frac{1}{2j-1} \le \int_{j-1}^{j} \frac{dx}{2x-1}$ for $j \ge 2$. [length 97/300] □

Definition 9 (Reversal transform)**.** Let $s \ge 1$ and $Q := g^s$. For $\ell, a \in \mathbb{Z}$ put $\Psi_\ell(a) := \sum_{m=0}^{Q-1} e(-(\ell \operatorname{rev}_s(m) + am)/Q)$, $M(\ell) := \max_{0\le a<Q} |\Psi_\ell(a)|$ and $S(s) := \sum_{0<|\ell|<Q} M(\ell)$. For $\theta \in \mathbb{R}$ let $f(\theta) := \frac{1}{g}\left|\sum_{d=0}^{g-1} e(d\theta)\right|$, and for $0 \le i < s$ let $\theta_i = \theta_i(\ell, a) := (\ell g^{s-1-i} + ag^i)/Q$ (Definitions 1 and 3).
Lean: `Psi`, `Mmax`, `Ssum`, `fdig`, `theta`

Lemma 10 (Periodicity)**.** *$\Psi_\ell(a)$ depends only on $\ell \bmod Q$ and $a \bmod Q$ (Definitions 1, 3 and 9).*
Lean: `Psi_periodic`

Proof. Changing ℓ or a by a multiple of Q changes each argument $-(\ell \operatorname{rev}_s(m) + am)/Q$ by an integer, as $\operatorname{rev}_s(m)$ and m are integers, and e has period 1. [length 113/300] □

Lemma 11 (Product formula)**.** *For all $\ell, a \in \mathbb{Z}$: $|\Psi_\ell(a)| = Q\prod_{i=0}^{s-1} f(\theta_i(\ell, a))$ (Definitions 1 and 9).*
Lean: `norm_Psi_eq`

Proof. For $0 \le m < Q$, $m = \sum_{i<s} m_i g^i$ (Definition 3 and Lemma 4), so $\ell \operatorname{rev}_s(m) + am = \sum_i m_i(\ell g^{s-1-i} + ag^i) = Q\sum_i m_i\theta_i$; and $(m_i)_{i<s}$ runs over $\{0, \dots, g-1\}^s$ as m does (Lemmas 2 and 4). So $\Psi_\ell(a) = \prod_{i<s}\sum_{d<g} e(-d\theta_i)$; each factor is the conjugate of $\sum_{d<g} e(d\theta_i)$, of modulus $gf(\theta_i)$. [length 204/300] □

Lemma 12 (Fourier inversion)**.** *For $\ell \in \mathbb{Z}$ and $0 \le n < Q$ (Definitions 1, 3 and 9):*

$$e(-\ell \operatorname{rev}_s(n)/Q) = Q^{-1}\sum_{a=0}^{Q-1} \Psi_\ell(a)e(an/Q).$$

Lean: `fourier_inversion`

Proof. By Definition 9 the right side is $Q^{-1}\sum_{m<Q} e(-\ell \operatorname{rev}_s(m)/Q)\sum_{a<Q} e(a(n-m)/Q)$. The inner sum is Q if $m = n$, and 0 otherwise, since it sums the powers of the Q-th root of unity $e((n-m)/Q) \ne 1$. [length 150/300] □

Lemma 13 (Zero frequency)**.** *If $Q \nmid \ell$, then $\Psi_\ell(0) = 0$ (Definitions 1, 3 and 9).*
Lean: `Psi_zero`

Proof. By Lemma 5, $n = \mathrm{rev}_s(m)$ runs over $\{0, \dots, Q-1\}$ as m does. So $\Psi_\ell(0) = \sum_{n<Q} \omega^n$ with $\omega := e(-\ell/Q)$, and $\omega \neq 1 = \omega^Q$; the sum is $(\omega^Q - 1)/(\omega - 1) = 0$. [length 107/300] □

Lemma 14 (Shape of f)**.** *f is continuous and 1-periodic, $0 \le f \le 1$, and $f(\theta) < 1$ for $\theta \notin \mathbb{Z}$ (Definitions 1 and 9).*
Lean: `fdig_shape`

Proof. Continuity and periodicity are clear, and $f \le 1$ by the triangle inequality. If $f(\theta) = 1$, equality holds there, so the unit vectors $e(d\theta)$, $d < g$, coincide; as $g \ge 2$, $e(\theta) = e(0) = 1$, i.e. $\theta \in \mathbb{Z}$. [length 152/300] □

Lemma 15 (Uniform gap)**.** *Let $\varepsilon_* := 1/(g^2(g+1))$ and $u_* := \left(\sup_{\varepsilon_* \le \theta \le 1-\varepsilon_*} f(\theta)\right)^{1/2}$. Then $0 \le u_* < 1$, and $f(\theta) \le u_*^2$ whenever $\|\theta\| \ge \varepsilon_*$ (Definitions 1 and 9).*
Lean: `epsStar`, `uStar`, `uStar_spec`

Proof. As $g \ge 2$, $0 < \varepsilon_* < 1/2$. By Lemma 14, f is continuous, ≥ 0, and < 1 on the compact set $[\varepsilon_*, 1-\varepsilon_*]$, which contains no integer; so the supremum is a maximum, and $0 \le u_*^2 < 1$. If $\|\theta\| \ge \varepsilon_*$ then $\theta - \lfloor\theta\rfloor \in [\varepsilon_*, 1-\varepsilon_*]$, and f has period 1. [length 173/300] □

Lemma 16 (Consecutive arguments)**.** *For $\ell, a \in \mathbb{Z}$ and $0 \le i \le s-2$: $g\theta_i - \theta_{i+1} = (g^2-1)\ell/g^{i+2}$, where $\theta_i = \theta_i(\ell, a)$ (Definition 9).*
Lean: `theta_consecutive`

Proof. By Definition 9, $g\theta_i = \ell/g^i + a/g^{s-1-i}$ and $\theta_{i+1} = \ell/g^{i+2} + a/g^{s-1-i}$. Subtracting gives $\ell/g^i - \ell/g^{i+2} = (g^2-1)\ell/g^{i+2}$. [length 96/300] □

Lemma 17 (One of two is far)**.** *For $0 \le i \le s-2$: $\max(\|\theta_i\|, \|\theta_{i+1}\|) \ge \|(g^2-1)\ell/g^{i+2}\|/(g+1)$ (Definitions 1 and 9).*
Lean: `one_of_two_far`

Proof. For $x, y \in \mathbb{R}$ and integers $n \ge 1$, $\|x-y\| \le \|x\| + \|y\|$ and $\|nx\| \le n\|x\|$. By Lemma 16, $\|(g^2-1)\ell/g^{i+2}\| = \|g\theta_i - \theta_{i+1}\| \le g\|\theta_i\| + \|\theta_{i+1}\|$, which is at most $(g+1)\max(\|\theta_i\|, \|\theta_{i+1}\|)$. [length 133/300] □

Definition 18 (Bad indices)**.** For $\ell \in \mathbb{Z}$ let $\ell' := (g^2-1)\ell \bmod Q$. For integers $n \ge 1$ and $r \ge 0$ with digits r_i, an index $0 \le i \le n-2$ is *bad* for r if (r_{i+1}, r_i) is neither $(0,0)$ nor $(g-1, g-1)$; let $\beta_n(r)$ be the number of these bad indices, and $b(\ell) := \beta_s(\ell')$ (Definitions 1, 3 and 9).
Lean: `twist`, `IsBad`, `badCount`, `bBad`

Lemma 19 (Twisting is bijective)**.** *The map $\ell \mapsto \ell'$ is a bijection of $\{0, \dots, Q-1\}$, and ℓ' (hence $b(\ell)$) depends only on $\ell \bmod Q$ (Definitions 1, 9 and 18).*
Lean: `twist_bijective`

Proof. A prime factor of $Q = g^s$ divides g^2, hence not $g^2 - 1$; so $\gcd(g^2-1, Q) = 1$, and multiplication by $g^2 - 1$ is invertible modulo Q. Also ℓ', and hence $b(\ell)$, depends only on $\ell \bmod Q$. [length 139/300] □

Lemma 20 (Fractional parts)**.** *For $0 \le i \le s-2$: $(g^2-1)\ell/g^{i+2} \equiv v_i + w_i \pmod 1$, where $v_i := \ell'_{i+1}/g + \ell'_i/g^2$ and $0 \le w_i < 1/g^2$ (Definitions 1, 3, 9 and 18).*
Lean: `frac_parts`

Proof. As $i+2 \le s$ and $(g^2-1)\ell \equiv \ell' \pmod{g^s}$, the number $((g^2-1)\ell - \ell')/g^{i+2}$ is an integer, and $\ell'/g^{i+2} \equiv (\ell' \bmod g^{i+2})/g^{i+2} = \sum_{j \le i+1} \ell'_j g^{j-i-2} \pmod 1$ (Lemma 4). This is $v_i + w_i$ with $w_i := \sum_{j<i} \ell'_j g^{j-i-2} < g^{-2}$. [length 159/300] □

Lemma 21 (Bad indices are far)**.** *If $0 \le i \le s-2$ is bad for ℓ', then $\|(g^2-1)\ell/g^{i+2}\| \ge 1/g^2$ (Definitions 1 and 18).*
Lean: `bad_far`

Proof. By Lemma 20 it suffices that $1/g^2 \le v_i + w_i \le 1 - 1/g^2$. Now v_i is a multiple of $1/g^2$ in $[0, 1-1/g^2]$; it is 0 only if $\ell'_{i+1} = \ell'_i = 0$, and $1-1/g^2$ only if both are $g-1$. So $1/g^2 \le v_i \le 1-2/g^2$, while $0 \le w_i < 1/g^2$. [length 157/300] □

Lemma 22 (Bad pair product)**.** *If $0 \le i \le s-2$ is bad for ℓ' (Definition 18), then $f(\theta_i) f(\theta_{i+1}) \le u_*^2$ for every $a \in \mathbb{Z}$ (Definition 9 and Lemma 15).*
Lean: `bad_pair_prod`

Proof. By Lemmas 17 and 21, $\max(\|\theta_i\|, \|\theta_{i+1}\|) \ge 1/(g^2(g+1)) = \varepsilon_*$ (Definition 1). By Lemma 15 that factor is at most u_*^2, and the other is at most 1 by Lemma 14. [length 122/300] □

Lemma 23 (Sup bound)**.** *For every $\ell \in \mathbb{Z}$: $M(\ell) \le Q\, u_*^{b(\ell)}$ (Definition 9, Definition 18 and Lemma 15).*
Lean: `Mmax_le`

Proof. Fix a and let $\varphi_i := f(\theta_i) \in [0,1]$ (Lemma 14). As $\varphi \le \varphi^{1/2}$ on $[0,1]$, $\prod_{i<s} \varphi_i \le \prod_{i \le s-2} (\varphi_i \varphi_{i+1})^{1/2}$, the right side being $(\varphi_0 \varphi_{s-1})^{1/2} \prod_{0<i<s-1} \varphi_i$ if $s \ge 2$. Use Lemma 22 for bad i, then Lemma 11. [length 147/300] □

Lemma 24 (Weighted digit strings)**.** *Let $0 \le u \le 1$ and $n \ge 1$. Then $\sum_{0 \le r < g^n} u^{\beta_n(r)} \le g(1+(g-1)u)^{n-1}$ (Definitions 3 and 18).*
Lean: `weighted_strings`

Proof. By Lemmas 2 and 4, $r \mapsto (r_i)_{i<n}$ maps $[0, g^n)$ bijectively onto $\{0, \dots, g-1\}^n$, and $\beta_n(r) =: \beta(d)$ depends only on $d = (r_i)_{i<n}$. Induct on n; for $n = 1$ the sum is g. Appending d_n to d raises β by 1 unless $d_n = d_{n-1} \in \{0, g-1\}$, so $\sum_{d_n} u^{\beta(d,d_n)} \le u^{\beta(d)}(1+(g-1)u)$, as $gu \le 1+(g-1)u$. [length 213/300] □

Lemma 25 (Sum over frequencies)**.** *Let $\rho := 1 + (g-1)u_*$. Then $\rho < g$, and $S(s) \le 2gQ\rho^{s-1}$ (Definition 9 and Lemma 15).*
Lean: `rho`, `Ssum_le`

Proof. $\rho < g$ as $u_* < 1$. By Lemma 10, each residue mod Q is ℓ mod Q for at most two ℓ with $0 < |\ell| < Q$, so $S(s) \le 2 \sum_{0 \le \ell < Q} M(\ell)$ (Definition 1). By Lemma 23, Lemma 19 and Definition 18 this is at most $2Q \sum_{0 \le r < Q} u_*^{\beta_s(r)}$, and Lemma 24 ($u = u_*$, $n = s$) applies. [length 185/300] □

Lemma 26 (Power saving)**.** *Let ρ be as in Lemma 25, $c := \min(1 - \log_g \rho, \frac{\log 2}{2 \log g})$ and $C := 2g$. Then $0 < c \le \frac{\log 2}{2 \log g}$, $C \ge 1$, and $S(s) \le C g^{(2-c)s}$ for all $s \ge 1$ (Definitions 1 and 9).*
Lean: `cSave`, `CSave`, `power_saving`

Proof. As $\rho < g$ (Lemma 25), $c > 0$. Also $1 \le \rho \le g^{1-c}$, as $u_* \ge 0$ (Lemma 15), so Lemma 25 gives $S(s) \le 2gQ\rho^{s-1} \le 2g \cdot g^s \cdot g^{(1-c)s} = C g^{(2-c)s}$; and $C = 2g \ge 1$. [length 108/300] □

Remark 27 (The size of c). Digit reversal is used in an essential way only in Lemmas 11 and 16, in which a cancels. The constant c of Lemma 26 is tiny; the accompanying paper [Pal] bounds $S(s)$ much more precisely, with a certified computation over windows of consecutive digits. Here only $c > 0$ matters.

Definition 28 (Palindromes)**.** For $L \geq 1$ let $\mathcal{P}_L$ be the set of integers $g^{L-1} \leq n < g^L$ whose digits n_i ($i < L$) satisfy $n_i = n_{L-1-i}$ for all $i < L$. Let $\mathcal{Q}_L := \{x \in \mathbb{N} : x^k \in \mathcal{P}_L\}$ and $X := g^{L/k}$ (Definitions 1 and 3).
Lean: `palSet`, `powSet`, `Xr`

Lemma 29 (Many palindromes)**.** *For $L \geq 1$: $\#\mathcal{P}_L \geq g^{L/2-1}$, and hence $X \leq (g\#\mathcal{P}_L)^{2/k}$ (Definitions 1 and 28).*
Lean: `many_palindromes`

Proof. Choose digits n_i for $(L-1)/2 \leq i < L$ with $n_{L-1} \neq 0$, and put $n_i := n_{L-1-i}$ for smaller i. Then $n := \sum_i n_i g^i \in [g^{L-1}, g^L)$ has these digits (Lemma 4), so it lies in $\mathcal{P}_L$, and distinct choices give distinct n (Lemma 2). So $\#\mathcal{P}_L \geq (g-1)g^{\lceil L/2 \rceil - 1} \geq g^{L/2-1}$, and $X = (g^{L/2})^{2/k}$. [length 208/300] □

Lemma 30 (First digits of palindromes)**.** *Let $n \in \mathcal{P}_L$ and $1 \leq t \leq L$. Then $g \nmid n$ and $\lfloor n/g^{L-t} \rfloor = \mathrm{rev}_t(n \bmod g^t)$ (Definitions 1, 3 and 28).*
Lean: `first_digits`

Proof. Let n_r be the digits of n. By Lemma 4, $n \bmod g^t = \sum_{r<t} n_r g^r$ has digits n_r, $r < t$, and $\lfloor n/g^{L-t} \rfloor = \sum_{r<t} n_{L-t+r} g^r$ as $n < g^L$. As $n_{L-t+r} = n_{t-1-r}$, the latter is $\mathrm{rev}_t(n \bmod g^t)$. For $t = 1$: $n \bmod g = \lfloor n/g^{L-1} \rfloor \geq 1$, as $n \geq g^{L-1}$. [length 162/300] □

Definition 31 (Nodes)**.** Let $1 \leq t \leq L$. A *node* (of level t) is an integer $0 \leq b < g^t$ with $g \nmid b^k$. For a node b put $T_b := \mathrm{rev}_t(b^k \bmod g^t)$, $y_0 := T_b g^{L-t}$, $y_1 := y_0 + g^{L-t}$, $I_b := [y_0^{1/k}, y_1^{1/k})$ (so an integer $x \geq 0$ lies in I_b iff $y_0 \leq x^k < y_1$) and $|I_b| := y_1^{1/k} - y_0^{1/k}$; let Z_b be the number of integers $x \in I_b$ with $x \equiv b \pmod{g^t}$, and if $Z_b \geq 1$ let x_b be the least of them. Let $Z(t) := Xg^{-2t}$ (Definitions 1, 3 and 28).
Lean: `IsNode`, `Tb`, `y0`, `y1`, `InI`, `Ilen`, `nodeElems`, `Zb`, `xb`, `Zt`

Lemma 32 (Leading digit)**.** *For a node b: $g^{t-1} \leq T_b < g^t$ (Definition 31).*
Lean: `leading_digit`

Proof. Let $n := b^k \bmod g^t$, with digits n_i (Definition 3). By Lemma 4, $n_0 = n \bmod g = b^k \bmod g \neq 0$, as $g \nmid b^k$ (Definition 1). So $T_b = \mathrm{rev}_t(n) \geq n_0 g^{t-1} \geq g^{t-1}$, and $T_b < g^t$ by Lemma 5. [length 130/300] □

Lemma 33 (The tree)**.** *Let $1 \leq t \leq L$ and $x \in \mathcal{Q}_L$. Then $b := x \bmod g^t$ is a node, $x \in I_b$, and $x = x_b + g^t z$ for an integer $0 \leq z < Z_b$ (Definitions 1, 28 and 31).*
Lean: `tree`

Proof. By Lemma 30 with $n = x^k$, $g \nmid x^k$, so $g \nmid b^k$ as $b \equiv x \pmod g$. Also $b^k \bmod g^t = x^k \bmod g^t$, so Lemma 30 gives $\lfloor x^k/g^{L-t} \rfloor = T_b$, i.e. $x \in I_b$. As $x \equiv b \pmod{g^t}$, x is one of the Z_b integers $x_b + g^t z$, $0 \leq z < Z_b$, in I_b. [length 150/300] □

Lemma 34 (Interval endpoints)**.** *For a node b: $g^{L-1} \leq y_0 < y_1 \leq \min(g^L, 2y_0)$, and $|I_b| = g^{L-t}/(k\xi^{1-1/k})$ for some $\xi \in [y_0, y_1]$ (Definition 31).*
Lean: `interval_endpoints`

Proof. By Lemma 32, $g^{t-1} \le T_b$ and $T_b+1 \le g^t$, which gives the inequalities, as $y_1/y_0 = 1+1/T_b \le 2$. The formula for $|I_b| = y_1^{1/k} - y_0^{1/k}$ is the mean value theorem for $y \mapsto y^{1/k}$. [length 131/300] □

Lemma 35 (Node sizes)**.** *For a node b: $Z(t)/k \le |I_b| g^{-t} \le gZ(t)/k$ and $Z(t)/k - 1 \le Z_b \le gZ(t)/k + 1$ (Definitions 28 and 31).*
Lean: `node_sizes`

Proof. By Lemma 34, $g^{(L-1)(1-1/k)} \le \xi^{1-1/k} \le g^{L(1-1/k)}$, so $|I_b|$ lies between $g^{L/k-t}/k = g^t Z(t)/k$ and $g^{L/k-t+1}/k$. An interval of length λ contains between $\lambda g^{-t} - 1$ and $\lambda g^{-t} + 1$ integers $\equiv b \pmod{g^t}$. [length 156/300] □

Lemma 36 (Nodes are not empty)**.** *Let $1 \le t \le L$ with $Z(t) \ge 2k$. Then every node b of level t has $1 \le Z_b \le 2gZ(t)/k$ (Definition 31).*
Lean: `nodes_nonempty`

Proof. By Lemma 35, $Z_b \ge Z(t)/k - 1 \ge 1$ and $Z_b \le gZ(t)/k + 1 \le 2gZ(t)/k$, as $gZ(t)/k \ge 1$. [length 60/300] □

Definition 37 (Phases)**.** Fix integers t, s with $1 \le s \le t-2$ and $2(t+s) \le L$; let $Q := g^s$ (as in Definition 9), $Z := Z(t)$ and $\Lambda := 1 + \log(1+6g^t)$. For a node b of level t put $\beta_b := kb^{k-1} \bmod Q$ and $d_b := \gcd(\beta_b, Q)$. If moreover $Z_b \ge 1$, then for $\ell, a \in \mathbb{Z}$ and an integer $z \ge 0$ put $\alpha_b := \lfloor x_b^k / g^t \rfloor \bmod Q$, $\delta_b := kx_b^{k-1} g^{2t+s-L}$, $\theta_{b,\ell,a} := (\ell\delta_b + a\beta_b)/Q$ and $n_b(z) := (\alpha_b + \beta_b z) \bmod Q$ (Definitions 1 and 31).
Lean: `Admissible`, `LambdaT`, `alphaB`, `betaB`, `dB`, `deltaB`, `thetaB`, `nB`

Definition 38 (Archimedean phases)**.** In the setting of Definition 37, for real $z \ge 0$ put $\phi_b(z) := (x_b + g^t z)^k / g^{L-t-s} - T_b g^s$ and $\lambda_b(z) := \phi_b(0) + \delta_b z$; and for real $T > 0$ put $\delta(T) := k(Tg^{L-t})^{1-1/k} g^{2t+s-L}$ (Definition 31).
Lean: `phiB`, `lambdaB`, `deltaT`

Lemma 39 (Formula for $\delta(T)$)**.** *For $T > 0$: $\delta(T) = \frac{kQ}{Z}(T/g^t)^{1-1/k}$; hence $kQ/(gZ) \le \delta(T) \le kQ/Z$ for $g^{t-1} \le T \le g^t$ (Definitions 28, 31, 37 and 38).*
Lean: `deltaT_formula`

Proof. As $Tg^{L-t} = (T/g^t)g^L$, $g^{L(1-1/k)} = g^L/X$ and $g^{2t}/X = 1/Z$,

$$\delta(T) = k\Big(\frac{T}{g^t}\Big)^{1-1/k} \frac{g^L}{X}\, g^{2t+s-L} = \frac{kQ}{Z}\Big(\frac{T}{g^t}\Big)^{1-1/k}.$$

Finally, if $T/g^t \in [1/g, 1]$ then $(T/g^t)^{1-1/k} \in [1/g, 1]$. [length 127/300] □

Lemma 40 (δ_b is near $\delta(T_b)$)**.** *For a node b of level t with $Z_b \ge 1$: $\delta(T_b) \le \delta_b \le (1 + g^{1-t})\delta(T_b)$ (Definitions 31, 37 and 38).*
Lean: `deltaB_near`

Proof. With y_0, y_1 as in Definition 31, $\delta_b/\delta(T_b) = (x_b/y_0^{1/k})^{k-1}$ and $x_b \in I_b = [y_0^{1/k}, y_1^{1/k})$. So the ratio lies in $[1, (y_1/y_0)^{1-1/k}] \subset [1, 1+1/T_b] \subset [1, 1+g^{1-t}]$, by Lemma 32. [length 136/300] □

Lemma 41 (Lower block)**.** *Let b be a node of level t with $Z_b \ge 1$, and $x = x_b + g^t z$ with an integer $z \ge 0$. Then $\lfloor x^k/g^t \rfloor \bmod Q = n_b(z)$ (Definitions 1, 31 and 37).*
Lean: `lower_block`

Proof. By the binomial theorem $x^k = x_b^k + kx_b^{k-1}g^t z + g^{2t}N$ with $N \in \mathbb{Z}$, so $\lfloor x^k/g^t \rfloor = \lfloor x_b^k/g^t \rfloor + kx_b^{k-1}z + g^t N$. Reduce modulo $Q = g^s$, using $s \le t$ and $x_b \equiv b \pmod{g^t}$. [length 120/300] □

Lemma 42 (Upper block)**.** *Let b be a node of level t with $Z_b \ge 1$, and $x = x_b + g^t z \in I_b$ with an integer $z \ge 0$. Then $\lfloor x^k/g^{L-t-s} \rfloor \bmod Q = \lfloor \phi_b(z) \rfloor$ (Definitions 1, 31, 37 and 38).*
Lean: `upper_block`

Proof. As $x \in I_b$, $T_b g^{L-t} \le x^k < (T_b+1)g^{L-t}$, so $\lfloor x^k/g^{L-t} \rfloor = T_b$. Hence $\lfloor x^k/g^{L-t-s} \rfloor \bmod g^s = \lfloor x^k/g^{L-t-s} \rfloor - g^s \lfloor x^k/g^{L-t} \rfloor = \lfloor x^k/g^{L-t-s} - T_b g^s \rfloor = \lfloor \phi_b(z) \rfloor$. [length 114/300] □

Lemma 43 (Taylor bound)**.** *Let b be a node of level t with $Z_b \ge 1$, $z \ge 0$ real and $x_b + g^t z \in I_b$. Then $0 \le \phi_b(z) - \lambda_b(z) \le \frac{k(k-1)}{2} y_1^{1-2/k} |I_b|^2 g^{t+s-L}$, with y_1 as in Definition 31 (Definitions 1, 31, 37 and 38).*
Lean: `taylor_bound`

Proof. ϕ_b is convex with $\phi_b'(0) = \delta_b$, so $\phi_b \ge \lambda_b$. By Taylor, $\phi_b(z) - \lambda_b(z) = \phi_b''(\zeta) z^2/2$ with $\zeta \in [0, z]$, and $\phi_b''(\zeta) z^2 = k(k-1)(x_b + g^t\zeta)^{k-2}(g^t z)^2 g^{t+s-L}$, where $x_b + g^t\zeta \le y_1^{1/k}$ and $g^t z \le |I_b|$. [length 148/300] □

Lemma 44 (Archimedean linearisation)**.** *In the setting of Lemma 43, $0 \le \phi_b(z) - \lambda_b(z) \le 1/g$ (Definitions 31, 37 and 38).*
Lean: `arch_linearisation`

Proof. By Lemma 34, $|I_b| \le g^{L-t}/(k y_0^{1-1/k})$, $y_1 \le 2y_0$ and $y_0 \ge g^{L-1}$. So the bound of Lemma 43 is at most $\frac{k-1}{2k}(y_1/y_0)^{1-2/k} y_0^{-1} g^{L-t+s} \le \frac{k-1}{k} g^{s+1-t} \le \frac{1}{g}$, as $s \le t-2$. [length 125/300] □

Lemma 45 (The palindrome condition)**.** *Let b be a node of level t with $Z_b \ge 1$, and $x = x_b + g^t z \in \mathcal{Q}_L \cap I_b$ with an integer $z \ge 0$. Then $\lambda_b(z) - \mathrm{rev}_s(n_b(z)) \in [-1/g, 1)$ (Definitions 3, 28, 31, 37 and 38).*
Lean: `palindrome_condition`

Proof. Let N_r be the digits of x^k. By Lemmas 41, 42 and 4, $n_b(z) = \sum_{i<s} N_{t+i} g^i$ and $\lfloor \phi_b(z) \rfloor = \sum_{i<s} N_{L-t-s+i} g^i$, which is $\mathrm{rev}_s(n_b(z))$ as $N_r = N_{L-1-r}$. Now use $0 \le \phi_b(z) - \lfloor \phi_b(z) \rfloor < 1$ and Lemma 44. [length 143/300] □

Lemma 46 (Fejér identity)**.** *For an integer $K \ge 1$ and $x \in \mathbb{R}$: $\sum_{|j|<K}(1 - |j|/K)e(jx) = K^{-1}\left|\sum_{i=0}^{K-1} e(ix)\right|^2$ (Definition 1).*
Lean: `fejer_identity`

Proof. The right side is $K^{-1}\sum_{i,i'<K} e((i-i')x)$, and for $|j| < K$ there are exactly $K - |j|$ pairs (i, i') with $0 \le i, i' < K$ and $i - i' = j$. [length 98/300] □

Lemma 47 (Fejér lower bound)**.** *If $K \ge 1$ is an integer and $|x| \le 1/(2K)$, then (Definition 1)*

$$K^{-1}\Big|\sum_{i<K} e(ix)\Big|^2 \ge 4K/\pi^2.$$

Lean: `fejer_lower`

Proof. For $x = 0$ the left side is K. Otherwise

$$\Big|\sum_{i<K} e(ix)\Big| = \frac{|\sin \pi K x|}{|\sin \pi x|} \geq \frac{2K|x|}{\pi|x|} = \frac{2K}{\pi},$$

as $\sin y \geq 2y/\pi$ for $0 \leq y \leq \pi/2$ and $|\sin y| \leq |y|$. [length 104/300] □

Lemma 48 (Size of K)**.** *For Q as in Definition 9, $K := \lfloor 2Q/3 \rfloor$ satisfies $Q/3 \leq K \leq 2Q/3 < Q$ and $K \geq 1$.*
Lean: `Kq`, `Kq_size`

Proof. $K \leq 2Q/3$ by definition, and $Q = g^s \geq 2$ (Definition 1). If $Q = 2$ then $K = 1 \geq 2/3$, and if $Q \geq 3$ then $K > 2Q/3 - 1 \geq Q/3 \geq 1$. [length 86/300] □

Definition 49 (Majorant)**.** With Q and K as in Lemma 48, let $F(y) := \frac{\pi^2}{4K^2}\big|\sum_{i=0}^{K-1} e(i(y - 1/4)/Q)\big|^2$ for $y \in \mathbb{R}$, and $\hat{F}_j := \frac{\pi^2}{4K}(1 - \frac{|j|}{K})e(-\frac{j}{4Q})$ for integers $|j| < K$ (Definitions 1 and 9).
Lean: `Fmaj`, `Fhat`

Lemma 50 (The majorant is large)**.** *$F \geq 0$, and $F(y) \geq 1$ for $y \in [-1/2, 1]$ (Definitions 1, 9 and 49).*
Lean: `maj_large`

Proof. $F \geq 0$ is clear. For $y \in [-1/2, 1]$, $x := (y - 1/4)/Q$ satisfies $|x| \leq 3/(4Q) \leq 1/(2K)$ by Lemma 48, so by Lemma 47, $F(y) = \frac{\pi^2}{4K} \cdot K^{-1}\big|\sum_{i<K} e(ix)\big|^2 \geq \frac{\pi^2}{4K} \cdot \frac{4K}{\pi^2} = 1$. [length 120/300] □

Lemma 51 (Majorant coefficients)**.** *$F(y) = \sum_{|j|<K} \hat{F}_j e(jy/Q)$ for all $y \in \mathbb{R}$, and $|\hat{F}_j| \leq 8/Q$ for $|j| < K$ (Definitions 1, 9 and 49).*
Lean: `maj_coeffs`

Proof. By Lemma 46 with $x = (y - 1/4)/Q$, $F(y) = \frac{\pi^2}{4K}\sum_{|j|<K}(1 - |j|/K)e(-j/(4Q))e(jy/Q) = \sum_{|j|<K} \hat{F}_j e(jy/Q)$. And $|\hat{F}_j| \leq \pi^2/(4K) \leq 3\pi^2/(4Q) < 8/Q$ by Lemma 48. [length 121/300] □

Lemma 52 (Counting in a node)**.** *For a node b of level t with $Z_b \geq 1$: $\#\{x \in \mathcal{Q}_L : x \equiv b \pmod{g^t}\} \leq \sum_{z=0}^{Z_b - 1} F(\lambda_b(z) - \mathrm{rev}_s(n_b(z)))$ (Definitions 3, 28, 31, 37, 38 and 49).*
Lean: `count_in_node`

Proof. By Lemma 33 each such x is $x_b + g^t z$ for some $0 \leq z < Z_b$, and lies in I_b; distinct x give distinct z. For these z the summand is ≥ 1 by Lemmas 45 and 50, as $[-1/g, 1) \subset [-1/2, 1]$ (Definition 1), and all summands are ≥ 0. [length 167/300] □

Lemma 53 (Twisted sums)**.** *For a node b of level t with $Z_b \geq 1$ and $0 < |j| < Q$ (Definitions 1, 3, 9, 31, 37 and 38):*

$$\Big|\sum_{z<Z_b} e\Big(\frac{j(\lambda_b(z) - \mathrm{rev}_s(n_b(z)))}{Q}\Big)\Big| \leq \frac{1}{Q}\sum_{0<a<Q} |\Psi_j(a)| \min\Big(Z_b, \frac{1}{2\|\theta_{b,j,a}\|}\Big).$$

Lean: `twisted_sum`

Proof. By Lemma 12 with $n = n_b(z) \equiv \alpha_b + \beta_b z$, and as $\lambda_b(z) = \lambda_b(0) + \delta_b z$, the sum is $Q^{-1}\sum_{a<Q} \Psi_j(a)e(\omega_a)\sum_{z<Z_b} e(z\theta_{b,j,a})$ with real ω_a. Apply Lemma 6; the term $a = 0$ vanishes by Lemma 13. [length 138/300] □

Definition 54 (Error term)**.** For a set $\mathcal{B}$ of nodes b of level t with $Z_b \geq 1$ let

$$E(\mathcal{B}) := \sum_{b\in\mathcal{B}} \sum_{0<|j|<Q} \sum_{0<a<Q} |\Psi_j(a)| \min\Big(Z_b, \frac{1}{2\|\theta_{b,j,a}\|}\Big)$$

(Definitions 1, 9, 31 and 37).
Lean: `Eerr`

Lemma 55 (Reduction)**.** *For $\mathcal{B}$ as in Definition 54: $\#\{x \in \mathcal{Q}_L : x \bmod g^t \in \mathcal{B}\} \leq \frac{8}{Q}\sum_{b\in\mathcal{B}} Z_b + \frac{8}{Q^2}E(\mathcal{B})$ (Definitions 1, 28 and 31).*
Lean: `reduction`

Proof. For $b \in \mathcal{B}$, by Lemmas 52 and 51 the count is at most $\sum_{|j|<K} |\hat{F}_j| |\sum_{z<Z_b} e(j(\lambda_b(z) - \mathrm{rev}_s(n_b(z)))/Q)|$ (Definitions 3, 37, 38 and 49). The term $j = 0$ is at most $8Z_b/Q$; bound the others by Lemma 53, as $K < Q$ (Lemma 48). Sum over $\mathcal{B}$. [length 175/300] □

Lemma 56 (Multiples)**.** *Let $Q \geq 1$ be any integer, $\beta \in \mathbb{Z}$ and $d := \gcd(\beta, Q)$. As a runs over $0, \ldots, Q-1$, $a\beta \bmod Q$ runs over the multiples rd, $0 \leq r < Q/d$, each exactly d times; equivalently, $\sum_{0\leq a<Q} \varphi(a\beta \bmod Q) = d\sum_{0\leq r<Q/d} \varphi(rd)$ for every function φ (Definition 1).*
Lean: `multiples_sum`

Proof. $a \mapsto a\beta \bmod Q$ is a homomorphism of $\mathbb{Z}/Q\mathbb{Z}$ whose image is generated by β, hence by $\gcd(\beta, Q) = d$; so the image is $d\mathbb{Z}/Q\mathbb{Z}$, of size Q/d, and each value is taken $Q/(Q/d) = d$ times. The second form follows by grouping the a by the value rd. [length 181/300] □

Lemma 57 (Distances)**.** *Let $R \geq 1$ and j be integers with $0 < |j| \leq R/2$, and let $|w_0| \leq 1/2$. Then $\|(j + w_0)/R\| \geq (2|j| - 1)/(2R)$ (Definition 1).*
Lean: `dist_lower`

Proof. For $n \in \mathbb{Z}$: if $n = 0$ then $|j + w_0 - nR| \geq |j| - 1/2$; if $n \neq 0$ then $|j + w_0 - nR| \geq R - |j| - 1/2 \geq |j| - 1/2$. Divide by R. [length 82/300] □

Lemma 58 (Sum over residues)**.** *Let $R \geq 1$ be an integer, $w \in \mathbb{R}$ and $A \geq 1$. Then $\sum_{r=0}^{R-1} \min(A, \frac{1}{2\|(w+r)/R\|}) \leq \min(A, \frac{R}{2\|w\|}) + R(2 + \log R)$ (Definition 1).*
Lean: `residue_sum`

Proof. Write $w = n_0 + w_0$ with $n_0 \in \mathbb{Z}$ and $|w_0| = \|w\| \leq 1/2$. As r runs mod R, $j \equiv n_0 + r$ runs over the integers in $(-R/2, R/2]$, and $\|(w + r)/R\| = \|(j + w_0)/R\|$. The term $j = 0$ is $\min(A, R/(2\|w\|))$; bound the others by Lemmas 57 and 8. [length 167/300] □

Lemma 59 (Sum over a)**.** *Let $y \in \mathbb{R}$, $\beta \in \mathbb{Z}$, $d := \gcd(\beta, Q)$ and $A \geq 1$. Then (Definitions 1 and 9)*

$$\sum_{a=0}^{Q-1} \min\Big(A, \frac{1}{2\|(y + a\beta)/Q\|}\Big) \leq \min\Big(dA, \frac{Q}{2\|y/d\|}\Big) + Q(2 + \log Q).$$

Lean: `a_sum`

Proof. By Lemma 56 the sum is $d\sum_{r<Q/d} \min(A, \frac{1}{2\|(y+rd)/Q\|})$, and $(y+rd)/Q = (y/d+r)/(Q/d)$. By Lemma 58 with $R = Q/d$ and $w = y/d$, it is at most $d\min(A, \frac{Q/d}{2\|y/d\|}) + d\frac{Q}{d}(2 + \log\frac{Q}{d})$. [length 137/300] □

Definition 60 (g-parts)**.** For a prime p let v_p be the p-adic valuation. For an integer $n \geq 1$ let $m_g(n) := \prod_{p|g} p^{v_p(n)}$, the product over the primes p dividing g. Let $\omega(g)$ be the number of primes dividing g, P the largest of them, and for real $D \geq 1$ let $\tau(D)$ be the number of integers $1 \leq m \leq D$ all of whose prime factors divide g (Definition 1).
Lean: `mg`, `omegaG`, `Pmax`, `tauG`

Lemma 61 (Few g-smooth numbers)**.** *τ is non-decreasing, and $\tau(D) \leq (1 + \log_2 D)^{\omega(g)}$ for $D \geq 1$ (Definition 60).*
Lean: `tauG_bound`

Proof. Monotonicity is clear. Each m counted is $\prod_{p|g} p^{e_p}$ with $p^{e_p} \leq D$, so $0 \leq e_p \leq \log_2 D$: at most $1 + \log_2 D$ choices for each of the $\omega(g)$ exponents. [length 109/300] □

Lemma 62 (Common factor of β_b)**.** *If b is a node of level t and $m := m_g(b)$, then d_b divides Q and km^{k-1} (Definitions 31, 37 and 60).*
Lean: `dB_dvd`

Proof. As $g \nmid b^k$, $b \neq 0$; write $b = mu$ with $\gcd(u, g) = 1$. Then $\beta_b \equiv km^{k-1}u^{k-1} \pmod{Q}$, so $d_b = \gcd(km^{k-1}u^{k-1}, Q) = \gcd(km^{k-1}, Q)$, as u is coprime to Q. [length 110/300] □

Lemma 63 (Binomial congruence)**.** *Let p be a prime, $j \geq 2$ and $u, w \in \mathbb{Z}$. Then $(u + p^j w)^k \equiv u^k + ku^{k-1}p^j w \pmod{p^{v_p(k)+j+1}}$ (Definition 60).*
Lean: `binomial_congr`

Proof. The other terms are $\binom{k}{i}u^{k-i}p^{ij}w^i$, $2 \leq i \leq k$. From $i\binom{k}{i} = k\binom{k-1}{i-1}$ and $v_p(i) \leq i - 1$ we get $v_p\binom{k}{i} \geq v_p(k) - i + 1$, and $ij - i + 1 \geq j + 1$, as $(i - 1)(j - 1) \geq 1$. [length 110/300] □

Lemma 64 (Close roots agree)**.** *Let p be a prime, $N > v_p(k)$, and $u, u' \in \mathbb{Z}$ with $p \nmid u$, $u^k \equiv u'^k \pmod{p^N}$ and $u \equiv u' \pmod{p^2}$. Then $u \equiv u' \pmod{p^{N-v_p(k)}}$ (Definition 60).*
Lean: `close_roots_agree`

Proof. If not, let j be maximal with $p^j \mid u' - u$; then $2 \leq j < N - v_p(k)$ and $u' = u + p^j w$ with $p \nmid w$. By Lemma 63, $u'^k - u^k \equiv ku^{k-1}p^j w \pmod{p^{v_p(k)+j+1}}$, whose valuation is $v_p(k) + j < N$, contradicting $p^N \mid u'^k - u^k$. [length 155/300] □

Lemma 65 (Few roots modulo p^N)**.** *Let p be a prime, $N \geq 1$ and $w \in \mathbb{Z}$ with $p \nmid w$. Then $u^k \equiv w \pmod{p^N}$ has at most $p^{v_p(k)+2}$ solutions $u \bmod p^N$ (Definitions 1 and 60).*
Lean: `roots_mod_prime_pow`

Proof. If $N \leq v_p(k) + 2$ this is clear. Otherwise every solution has $p \nmid u$, and by Lemma 64, solutions in the same class mod p^2 agree mod $p^{N-v_p(k)}$. So each of the p^2 classes contains at most $p^{v_p(k)}$ solutions mod p^N. [length 165/300] □

Lemma 66 (Few roots modulo n)**.** *Let $n \geq 1$ have all its prime factors dividing g, and $\gcd(w, n) = 1$. Then $u^k \equiv w \pmod{n}$ has at most kg^2 solutions $u \bmod n$ (Definitions 1 and 60).*
Lean: `roots_mod_n`

Proof. By the Chinese remainder theorem the number of solutions is the product over $p \mid n$ of the numbers of solutions mod $p^{v_p(n)}$, each at most $p^{v_p(k)+2}$ by Lemma 65. Finally $\prod_{p|g} p^{v_p(k)} \leq k$ and $\prod_{p|g} p^2 \leq g^2$. [length 158/300] □

Lemma 67 (Rigid g-part)**.** *Let $D \geq 1$ with $D^k < 2^t$, and let b, b' be nodes of level t with $T_b = T_{b'}$ and $m := m_g(b) \leq D$ (Definitions 31 and 60). Then $m^k \mid g^t$ and $m_g(b') = m$.*
Lean: `rigid_gpart`

Proof. Let $p \mid g$ and $v := v_p(b)$. Then $p^{kv} \leq m^k < 2^t \leq p^{v_p(g)t}$, so $kv < v_p(g)t$; hence $m^k \mid g^t$. By Lemma 5, $b^k \equiv b'^k \pmod{g^t}$, so $b'^k \equiv b^k \pmod{p^{v_p(g)t}}$, $v_p(b'^k) = kv$ and $v_p(b') = v$. [length 129/300] □

Lemma 68 (Fibres)**.** *Let $D \geq 1$ with $D^k < 2^t$, and b a node of level t with $m := m_g(b) \leq D$. Then at most kg^2m^{k-1} nodes b' of level t satisfy $T_{b'} = T_b$ (Definitions 31 and 60).*
Lean: `fibre_bound`

Proof. By Lemma 67, $m^k \mid g^t$, $b = mu$, $b' = mu'$ with u, u' coprime to g. By Lemma 5, $b'^k \equiv b^k \pmod{g^t}$, so $u'^k \equiv u^k \pmod{n}$, $n := g^t/m^k$. By Lemma 66, u' lies in at most kg^2 classes mod n, each with m^{k-1} lifts mod g^t/m, which fix b'. [length 168/300] □

Remark 69 (The archimedean count). The next blocks bound the sums over a and over the nodes. The g-adic phases $a\beta_b$ are summed for each node separately (Lemma 59). The averaging over nodes comes from the archimedean phases $\ell\delta_b$: they are close to the values of the monotone function $\ell\delta(T)$ at $T = T_b$, these values are well spaced, and few nodes share the same T_b (Lemma 68).

Definition 70 (Data for one frequency)**.** In the setting of Definition 37, let $0 < |\ell| < Q$ and $D \geq 1$ with $D^k < 2^t$, and suppose $Z \geq 2k$, so that every node b of level t has $Z_b \geq 1$ (Lemma 36). Let $\mathcal{B}$ be a set of nodes b of level t with $m_g(b) \leq D$ (Definitions 31 and 60). Let $Z_+ := gZ/k + 1$, and for $b \in \mathcal{B}$ let $y_b := \ell\delta_b/d_b$, $\Phi_b := \min(d_bZ_+, \frac{Q}{2\|y_b\|})$ and $W_\ell := \sum_{b\in\mathcal{B}} \sum_{0<a<Q} \min(Z_b, \frac{1}{2\|\theta_{b,\ell,a}\|})$ (Definition 1).
Lean: `FreqSetting`, `Zplus`, `yB`, `PhiB`, `Wl`

Lemma 71 (From W_ℓ to Φ_b)**.** $W_\ell \leq \sum_{b\in\mathcal{B}} \Phi_b + g^tQ(2 + \log Q)$ *(Definitions 1, 37 and 70).*
Lean: `W_to_Phi`

Proof. By Lemmas 36 and 35, $1 \leq Z_b \leq Z_+$. For each $b \in \mathcal{B}$ apply Lemma 59 with $y = \ell\delta_b$, $\beta = \beta_b$ and $A = Z_b$, after adding the term $a = 0$, and use $\min(d_bZ_b, \cdot) \leq \min(d_bZ_+, \cdot)$. Finally $\#\mathcal{B} \leq g^t$ (Definition 31). [length 147/300] □

Lemma 72 (Small phases)**.** *If $b \in \mathcal{B}$ and $|\ell|\delta_b \leq d_b/2$, then $\Phi_b \leq gZd_b/(2k|\ell|)$ (Definitions 1, 37 and 70).*
Lean: `small_phases`

Proof. Then $|y_b| \leq 1/2$, so $\|y_b\| = |\ell|\delta_b/d_b$ and $\Phi_b \leq Qd_b/(2|\ell|\delta_b)$. By Lemmas 40, 39 and 32, $\delta_b \geq \delta(T_b) \geq kQ/(gZ)$ (Definitions 31 and 38). [length 103/300] □

Lemma 73 (Sum of d_b)**.** $\sum_{b\in\mathcal{B}} d_b \leq k\tau(D)D^{k-2}g^t$ *(Definitions 1, 37, 60 and 70).*
Lean: `sum_dB`

Proof. For $b \in \mathcal{B}$ let $m := m_g(b) \leq D$. By Lemma 62, $d_b \leq km^{k-1}$, and $m \mid g^t$ by Lemma 67 (with $b' = b$). Exactly g^t/m integers in $[0, g^t)$ are divisible by m, and m takes at most $\tau(D)$ values; so the sum is at most $\tau(D) \max_m (g^t/m)km^{k-1} \leq k\tau(D)D^{k-2}g^t$, as $k \geq 2$. [length 189/300] □

Definition 74 (Phase curve)**.** In the setting of Definition 70, for an integer $d \geq 1$ and real $T > 0$ let $h_d(T) := \ell\delta(T)/d$; let $\mathcal{T}_d$ be the set of integers $T \in [g^{t-1}, g^t)$ with $|h_d(T)| \geq 1/3$, and put $\mu := (1-1/k)|h_d(g^t)|/g^t$ and $\Delta := |h_d(g^t) - h_d(g^{t-1})|$ (both depend on d) (Definitions 1 and 38).
Lean: `hd`, `Tset`, `muD`, `DeltaD`

Lemma 75 (Perturbation)**.** *If $b \in \mathcal{B}$ and $|\ell|\delta_b > d_b/2$, then $T_b \in \mathcal{T}_{d_b}$ and $|y_b - h_{d_b}(T_b)| \leq \varepsilon_b := kg^{s-1}/(d_b Z)$ (Definitions 1, 31, 37, 38, 70 and 74).*
Lean: `epsB`, `perturbation`

Proof. Let $h := h_{d_b}(T_b)$. By Lemmas 40, 39 and 32, $|y_b - h| \leq |\ell|g^{1-t}\delta(T_b)/d_b \leq g^s g^{1-t} kQ/(Zd_b) \leq \varepsilon_b$, as $s \leq t-2$. Also $|h| \geq |\ell|\delta_b/((1+g^{1-t})d_b) > 1/(2+2g^{1-t}) \geq 1/3$, as $t \geq 3$; and $g^{t-1} \leq T_b < g^t$. [length 145/300] □

Lemma 76 (Replacement)**.** *In the setting of Lemma 75, $\Phi_b \leq 5\min(d_b Z_+, \frac{Q}{2\|h_{d_b}(T_b)\|})$ (Definitions 1, 31, 37, 70 and 74).*
Lean: `replacement`

Proof. Let $h := h_{d_b}(T_b)$. If $\|h\| \geq 2\varepsilon_b$ then $\|y_b\| \geq \|h\|/2$, which gives the factor 2. Otherwise $\frac{Q}{2\|h\|} > \frac{Q}{4\varepsilon_b} = \frac{gd_bZ}{4k}$ and $d_bZ_+ \geq \frac{gd_bZ}{k}$, so the right side is at least $\frac{5gd_bZ}{4k} \geq d_bZ_+ \geq \Phi_b$, as $gZ \geq 4k$. [length 146/300] □

Lemma 77 (Shape of h_d)**.** *On $[g^{t-1}, g^t]$, $|h_d|$ is non-decreasing, $h_d'(T) = (1-1/k)h_d(T)/T$, $|h_d'|$ is non-increasing, and $|h_d'(T)| \leq g\mu$, where $\mu = |h_d'(g^t)|$ (Definitions 1, 37 and 74).*
Lean: `hd_shape`

Proof. By Lemma 39, $h_d(T) = \frac{\ell kQ}{dZ}(T/g^t)^{1-1/k}$. This gives the first two claims, and $|h_d'(T)|$ is a constant times $T^{-1/k}$, so it is non-increasing, $|h_d'(g^t)| = (1-\frac{1}{k})|h_d(g^t)|/g^t = \mu$, and $|h_d'(T)|/\mu = (g^t/T)^{1/k} \leq g$. [length 170/300] □

Lemma 78 (Gaps)**.** *Suppose $\mathcal{T}_d \neq \emptyset$. Then $\mu \geq 1/(6g^t)$, and $|h_d(T') - h_d(T)| \geq \mu$ for distinct $T, T' \in \mathcal{T}_d$ (Definitions 1 and 74).*
Lean: `gaps`

Proof. By Lemma 77, $|h_d(g^t)| \geq |h_d(T)| \geq 1/3$ for $T \in \mathcal{T}_d$, so $\mu = (1-\frac{1}{k})|h_d(g^t)|/g^t \geq 1/(6g^t)$. Also $|h_d'| \geq \mu$ on $[g^{t-1}, g^t]$, and $|T' - T| \geq 1$; apply the mean value theorem. [length 128/300] □

Lemma 79 (Range)**.** *If $\mathcal{T}_d \neq \emptyset$, then $h_d(T)$ lies between $h_d(g^{t-1})$ and $h_d(g^t)$ for every $T \in \mathcal{T}_d$; moreover $\Delta \leq g^{t+1}\mu$ and $\Delta \leq kg^{2s}/(dZ)$ (Definitions 37, 70 and 74).*
Lean: `range_bound`

Proof. By Lemma 39, $h_d(T) = \frac{\ell kQ}{dZ}(T/g^t)^{1-1/k}$ is monotone and of constant sign on $[g^{t-1}, g^t]$. This gives the first claim, and $\Delta \leq |h_d(g^t)| \leq |\ell|kQ/(dZ) \leq kg^{2s}/(dZ)$, as $|\ell| < Q = g^s$. By Lemma 77 and the mean value theorem, $\Delta \leq g^t \cdot g\mu$. [length 180/300] □

Lemma 80 (Spacing near one integer)**.** *Let $\mu > 0$, $m \geq 1$, $A, H > 0$ be real and $n \in \mathbb{Z}$. Let finitely many reals $y_i \in (n - \frac{1}{2}, n + \frac{1}{2}]$ with integer weights $w_i \geq 0$ be such that every half-open interval $[a, a+\mu)$ carries total weight $\sum_{y_i \in [a,a+\mu)} w_i \leq m$. Then $\sum_i w_i \min(A, \frac{H}{|y_i - n|}) \leq 2mA + 2m\frac{H}{\mu}(1 + \log(1 + \frac{1}{\mu}))$ (Definition 1).*
Lean: `spacing_one`

Proof. Let $J := \lfloor 1/(2\mu) \rfloor + 1$. The intervals $[n + j\mu, n + (j+1)\mu)$ and $[n - (j+1)\mu, n - j\mu)$, $0 \le j < J$, cover $(n - \frac{1}{2}, n + \frac{1}{2}]$, each carries weight at most m, and the y_i in them have $|y_i - n| \ge j\mu$. Use $\sum_{1 \le j < J} \frac{1}{j} \le 1 + \log(1 + \frac{1}{\mu})$ (Lemma 7). [length 170/300] □

Lemma 81 (Spacing sum). *Let $\mu > 0$, $m \ge 1$, $A, H > 0$, $\Delta \ge 0$ and a_0 be real. Let finitely many reals $y_i \in [a_0, a_0 + \Delta]$ with integer weights $w_i \ge 0$ be such that every half-open interval of length μ carries total weight at most m. Then $\sum_i w_i \min(A, \frac{H}{\|y_i\|}) \le m(\Delta + 2)(2A + 2\frac{H}{\mu}(1 + \log(1 + \frac{1}{\mu})))$ (Definition 1).*
Lean: `spacing_sum`

Proof. Each y_i lies in $(n - \frac{1}{2}, n + \frac{1}{2}]$ for a unique $n \in \mathbb{Z}$, and then $\|y_i\| = |y_i - n|$. These n lie in $[a_0 - \frac{1}{2}, a_0 + \Delta + \frac{1}{2})$, so at most $\Delta + 2$ of them occur. Apply Lemma 80 for each n. [length 127/300] □

Lemma 82 (Count over T). *Let $d \ge 1$ be an integer with $\mathcal{T}_d \ne \emptyset$, let $\kappa \ge 1$ be real, and let integers $0 \le N_T \le \kappa$ be given for $T \in \mathcal{T}_d$. Then (Definitions 1, 37, 70 and 74)*

$$\sum_{T \in \mathcal{T}_d} N_T \min\Big(dZ_+, \frac{Q}{2\|h_d(T)\|}\Big) \le \kappa(\Delta + 2)\Big(2dZ_+ + \frac{Q}{\mu}\Lambda\Big).$$

Lean: `count_over_T`

Proof. Apply Lemma 81 to the numbers $h_d(T)$, $T \in \mathcal{T}_d$, with weights N_T, $m = \kappa$, $A = dZ_+$, $H = Q/2$, and the interval between $h_d(g^{t-1})$ and $h_d(g^t)$ (Lemma 79): by Lemma 78, $\mu > 0$, each half-open interval of length μ contains $h_d(T)$ for at most one T, and $1 + \log(1 + 1/\mu) \le \Lambda$. [length 202/300] □

Lemma 83 (Count over T, explicitly). *If moreover $d \le kD^{k-1}$, the bound in Lemma 82 is at most $\kappa(4g^{2s+1} + 8gD^{k-1}Z + 7g^{t+1}Q\Lambda)$ (Definitions 1, 37, 70 and 74).*
Lean: `count_over_T_explicit`

Proof. By Lemma 79, $dZ_+\Delta \le \frac{2gZ}{k} \cdot \frac{kg^{2s}}{Z} = 2g^{2s+1}$, as $Z_+ \le 2gZ/k$ (since $Z \ge 2k$). Next $4dZ_+ \le 8gD^{k-1}Z$. Finally $(\Delta+2)/\mu \le g^{t+1} + 2/\mu \le g^{t+1} + 12g^t \le 7g^{t+1}$ by Lemma 78. [length 121/300] □

Lemma 84 (Large phases). *Let $\mathcal{B}_2 := \{b \in \mathcal{B} : |\ell|\delta_b > d_b/2\}$. Then (Definitions 1, 31, 37, 60 and 70)*

$$\sum_{b \in \mathcal{B}_2} \Phi_b \le 5\tau(kD^{k-1})kg^2D^{k-1}(4g^{2s+1} + 8gD^{k-1}Z + 7g^{t+1}Q\Lambda).$$

Lean: `large_phases`

Proof. By Lemmas 75 and 76, $\Phi_b \le 5\min(d_bZ_+, \frac{Q}{2\|h_{d_b}(T_b)\|})$, $T_b \in \mathcal{T}_{d_b}$ (Definition 74). By Lemma 62, $d_b \mid Q$ and $d_b \le kD^{k-1}$: at most $\tau(kD^{k-1})$ values. For each value d use Lemma 83 with $\kappa := kg^2D^{k-1}$ and $N_T := \#\{b \in \mathcal{B}_2 : d_b = d, T_b = T\} \le \kappa$ (Lemma 68). [length 183/300] □

Lemma 85 (Bound for W_ℓ). *In the setting of Definition 70 (with Definitions 1, 37 and 60),*

$$W_\ell \le g^tQ(2 + \log Q) + \frac{g\tau(D)D^{k-2}Zg^t}{2|\ell|}$$
$$+5\tau(kD^{k-1})kg^2D^{k-1}(4g^{2s+1} + 8gD^{k-1}Z + 7g^{t+1}Q\Lambda).$$

Lean: `W_bound`

Proof. By Lemma 71 it suffices to bound $\sum_{b\in\mathcal{B}}\Phi_b$. For $b\in\mathcal{B}\setminus\mathcal{B}_2$ use Lemma 72 and then Lemma 73: $\sum\Phi_b\le\frac{gZ}{2k|\ell|}k\tau(D)D^{k-2}g^t$. For $b\in\mathcal{B}_2$ use Lemma 84. [length 106/300] □

Lemma 86 (Bound for E)**.** *In the setting of Definition 37, let $D\ge1$ with $D^k<2^t$, suppose $Z\ge2k$, and let $\mathcal{B}$ be the set of all nodes b of level t with $m_g(b)\le D$, so that $Z_b\ge1$ for $b\in\mathcal{B}$ (Lemma 36). Then $E(\mathcal{B})\le S(s)V+(1+\log Q)Qg\tau(D)D^{k-2}Zg^t$, where V is the sum of the first and last terms in Lemma 85 (Definitions 1, 9, 31, 54 and 60).*
Lean: `nodesSmall`, `Vterm`, `E_bound`

Proof. As $|\Psi_\ell(a)|\le M(\ell)$, $E(\mathcal{B})\le\sum_{0<|\ell|<Q}M(\ell)W_\ell$, with W_ℓ as in Definition 70. Insert Lemma 85. In the middle term use $M(\ell)\le Q$ (each term of $\Psi_\ell(a)$ has modulus 1) and $\sum_{0<|\ell|<Q}\frac{1}{2|\ell|}\le1+\log Q$ (Lemma 7). [length 154/300] □

Lemma 87 (A divisor in a window)**.** *Let $D\ge1$ and $n\ge1$ with $m_g(n)>D$. Then $m_g(n)$ has a divisor m' with $D<m'\le PD$ (Definition 60).*
Lean: `divisor_window`

Proof. Write $m_g(n)=p_1\cdots p_r$ with primes $p_i\mid g$, so $p_i\le P$, and let $m_j:=p_1\cdots p_j$. Then $m_0=1\le D<m_r$; for the least j with $m_j>D$ we have $m_j\le Pm_{j-1}\le PD$. [length 110/300] □

Lemma 88 (A good level)**.** *Let $L\ge1$, $D\ge1$ with $PD\le X^{\log 2/(2\log g)}$, and $t':=\lceil L/(2k)\rceil$. Then $X^{1/2}\le g^{t'}<gX^{1/2}$, $1\le t'\le L$, $Z(t')\le1$, and every divisor $m'\le PD$ of some $m_g(n)$, $n\ge1$, divides $g^{t'}$ (Definitions 1, 28, 31 and 60).*
Lean: `tPrime`, `good_level`

Proof. As $X^{1/2}=g^{L/(2k)}$ and $L/(2k)\le t'<L/(2k)+1$, the first claim holds; $1\le t'\le L$ as $0<L/(2k)\le L$; and $Z(t')=Xg^{-2t'}\le1$. For $p\mid g$, $p^{v_p(m')}\le PD\le(X^{1/2})^{\log 2/\log g}\le2^{t'}\le p^{v_p(g)t'}$; and m' has no other prime factors. [length 167/300] □

Lemma 89 (Large g-part)**.** *Under the hypotheses of Lemma 88, $\#\{x\in\mathcal{Q}_L:m_g(x)>D\}\le\tau(PD)(g/k+1)gX^{1/2}/D$ (Definitions 1, 28, 31 and 60).*
Lean: `large_gpart`

Proof. For such x pick m' by Lemma 87; then $m'\mid g^{t'}$ (Lemma 88), so m' divides the node $b:=x \bmod g^{t'}$ (Lemma 33). For fixed m' there are $g^{t'}/m'<gX^{1/2}/D$ such b, each containing at most $Z_b\le g/k+1$ such x (Lemmas 33 and 35); and m' takes at most $\tau(PD)$ values. [length 197/300] □

Definition 90 (Parameters)**.** Let c,C be as in Lemma 26, $\eta:=c/(32k)$, $\nu:=(2-c)/(4-c)$, $D:=X^\eta$, $t:=\lfloor L/(k(4-c))\rfloor$ $(=\lfloor\log_g X/(4-c)\rfloor)$ and $s:=t-2$ (Definitions 1 and 28).
Lean: `eta`, `nu`, `Dpar`, `tpar`, `spar`

Lemma 91 (Admissible parameters)**.** *For L large: $t\ge3$, $1\le s\le t-2$, $2(t+s)\le L$, $Z(t)\ge2k$, $g^t\le X^{1/(4-c)}<g^{t+1}$, and $Z(t)\le g^2X^\nu$ (Definitions 1, 28, 31 and 90).*
Lean: `admissible_params`

Proof. The fifth claim holds by the definition of t, as $X^{1/(4-c)}=g^{L/(k(4-c))}$. As $\log_g X=L/k$ and $k(4-c)\ge6$ (Lemma 26), $t\ge L/(4k)-1$ and $t\le L/6$, so $t\ge3$ for L large, and $2(t+s)\le4t\le L$. Next, $Z(t)=Xg^{-2t}$ lies between $X^{1-2/(4-c)}=X^\nu$ and g^2X^ν, and $X^\nu\to\infty$, as $c<2$. [length 202/300] □

Lemma 92 (Admissible D)**.** *For L large: $D^k<2^t$ and $PD\le X^{\log 2/(2\log g)}$ (Definitions 1, 28, 60 and 90).*
Lean: `admissible_D`

Proof. By Lemma 26, $c \le \frac{\log 2}{2\log g}$, so $D^k = X^{c/32} \le X^{\log 2/(64\log g)}$ and $PD \le PX^{\log 2/(64\log g)}$. By Lemma 91, $2^t > 2^{-1}\cdot 2^{\log_g X/(4-c)} \ge 2^{-1}X^{\log 2/(4\log g)}$. Both claims follow for L large. [length 142/300] □

Definition 93 (The main set of nodes)**.** Let L be large enough for Lemmas 91 and 92. Then t and s (Definition 90) satisfy the hypotheses of Definition 37, whose notation (Q, $Z = Z(t)$, Λ, ...) we use for them, and $D^k < 2^t$. Let $\mathcal{B}$ be the set of nodes b of level t with $m_g(b) \le D$ (Definitions 31 and 60).
Lean: `Bmain`

Lemma 94 (Main term)**.** *For L large: $\frac{8}{Q}\sum_{b\in\mathcal{B}} Z_b \ll X^\nu$ (Definitions 1, 28, 31, 90 and 93).*
Lean: `main_term`

Proof. By Lemma 91, $Z \ge 2k$, so Lemma 36 gives $\sum_{b\in\mathcal{B}} Z_b \le g^t \cdot 2gZ/k$. Now $g^t/Q = g^2$ and $Z \le g^2X^\nu$ by Lemma 91. [length 73/300] □

Lemma 95 (Auxiliary bounds)**.** *For L large: $S(s) \le CQ^2$, $S(s) \le CX^\nu$, $g^t = g^2Q$, $2+\log Q \le \Lambda \ll L$, and $\tau(D) \le \tau(kD^{k-1}) \ll L^{\omega(g)}$ (Definitions 1, 9, 28, 60, 90 and 93).*
Lean: `aux_bounds`

Proof. By Lemma 26, $S(s) \le Cg^{(2-c)s}$, which is at most $Cg^{2s} = CQ^2$ and at most $Cg^{(2-c)t} \le CX^\nu$ (Lemma 91). As $g^t = g^2Q \ge 4Q$, $2+\log Q \le 1+\log(24Q) \le \Lambda \le 1+\log(7g^t) \ll L$. By Lemma 61, as $X = g^{L/k}$ and τ is non-decreasing,

$$\tau(kD^{k-1}) \le (1+\log_2(kX^{k\eta}))^{\omega(g)} \ll L^{\omega(g)}.$$

[length 194/300] □

Lemma 96 (First error piece)**.** *For L large: $S(s)V/Q^2 \ll L^{\omega(g)+1}D^{2k-2}X^\nu$, with V as in Lemma 86 for the parameters of Definition 93 (Definitions 1, 9, 28, 60 and 90).*
Lean: `err_piece1`

Proof. Write $S = S(s)$. By Lemmas 85, 95 and 91,

$$Sg^tQ(2+\log Q)/Q^2 \le g^2S\Lambda \ll LX^\nu, \qquad Sg^{2s+1}/Q^2 = gS \ll X^\nu,$$
$$SD^{k-1}Z/Q^2 \le CD^{k-1}Z \ll D^{k-1}X^\nu, \qquad Sg^{t+1}Q\Lambda/Q^2 = g^3S\Lambda \ll LX^\nu.$$

Multiply the last three by $\tau(kD^{k-1})D^{k-1} \ll L^{\omega(g)}D^{k-1}$. [length 161/300] □

Lemma 97 (Second error piece)**.** *For L large (Definitions 1, 28, 60, 90 and 93):*

$$(1+\log Q)Qg\tau(D)D^{k-2}Zg^t/Q^2 \ll L^{\omega(g)+1}D^{k-2}X^\nu.$$

Lean: `err_piece2`

Proof. By Lemma 95, $g^t/Q = g^2$, $1+\log Q \le \Lambda \ll L$ and $\tau(D) \ll L^{\omega(g)}$; and $Z \le g^2X^\nu$ by Lemma 91. [length 61/300] □

Lemma 98 (Small g-part)**.** *For L large: $\#\{x \in \mathcal{Q}_L : m_g(x) \le D\} \ll L^{\omega(g)+1}D^{2k-2}X^\nu$ (Definitions 1, 28, 31, 60, 90 and 93).*
Lean: `small_gpart`

Proof. If $m_g(x) \le D$ then $p^{v_p(x)} \le D < 2^t \le p^{v_p(g)t}$ for $p \mid g$, so $b := x \bmod g^t$ has $m_g(b) = m_g(x)$, and b is a node (Lemma 33); so $b \in \mathcal{B}$. By Lemmas 91 and 36, $Z_b \ge 1$ on $\mathcal{B}$. Apply Lemma 55, then Lemmas 94, 86, 96 and 97. [length 153/300] □

Lemma 99 (Exponents)**.** $\nu + (2k-2)\eta \le \frac{1}{2} - \eta$ *(Definition 90).*
Lean: `exponents`

Proof. By Lemma 26, $0 < c < 4$, and $\nu = \frac{1}{2} - \frac{c}{2(4-c)} \le \frac{1}{2} - \frac{c}{8} = \frac{1}{2} - 4k\eta$; and $(2k-2)\eta + \eta < 2k\eta \le 4k\eta$. [length 66/300] □

Lemma 100 (Counting solutions)**.** *For L large: $\#\mathcal{Q}_L \ll X^{(1-\eta)/2}$ (Definitions 1, 28, 60 and 90).*
Lean: `count_solutions`

Proof. Lemma 89 applies with $D = X^\eta$ (by Lemma 92) and gives $\ll \tau(PD)X^{1/2-\eta} \ll L^{\omega(g)}X^{1/2-\eta}$ (Lemma 61). By Lemmas 98 and 99, $\#\{x : m_g(x) \le D\} \ll L^{\omega(g)+1}X^{1/2-\eta}$. Finally $L^{\omega(g)+1} \le X^{\eta/2}$ for L large, as $\eta > 0$ (Lemma 26). [length 162/300] □

Theorem 101 (Palindromic perfect powers)**.** *For all integers $g \ge 2$ and $k \ge 2$ there are $\gamma = \gamma(g,k) \in (0,1)$ and $C' = C'(g,k)$ such that, for every $L \ge 1$, the number of L-digit palindromes in base g that are perfect k-th powers is at most $C'(\#\mathcal{P}_L)^{\gamma/k}$ (Definitions 1 and 28).*
Lean: `main_theorem`, `main_theorem_unconditional`, `main_theorem_digits`

Proof. The map $x \mapsto x^k$ is a bijection from $\mathcal{Q}_L$ onto these powers. Let $\gamma := 1 - \eta \in (0,1)$, as $0 < c < 1$ (Definition 90 and Lemma 26). For L large, Lemmas 100 and 29 give $\#\mathcal{Q}_L \ll X^{(1-\eta)/2} \le (g\#\mathcal{P}_L)^{\gamma/k}$. For the finitely many other L, $\#\mathcal{Q}_L \le \#\mathcal{P}_L$; enlarge C'. [length 186/300] □

Remark 102 (Comparison). Cilleruelo, Luca and Shparlinski [CLS] proved Theorem 101 with $\gamma = 1$. The value $\gamma = 1 - c/(32k)$ obtained here is very close to 1, as c is tiny. The accompanying paper [Pal], with sharper bookkeeping and a certified computation of $S(s)$, gives for instance $\gamma = 0.9674$ for $g = 2$ and every k.

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